Can the Poona Dictionary Be Finished? Citation Density, Editorial Scale, and the Coverage of Its Cited Sources

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Abstract

The Poona Dictionary (PD) began in 1948 with a thirty-seven-volume plan and ten million quotation slips. By 2019 it had issued thirty-five fascicle-volumes and 6,056 pages, all within letter *a*. This paper explains that outcome through two linked processes: the workload created by PD's lemmatisation and citation choices, and the editorial conversion of quotation slips into finished entries. A same-unit comparison with the completed Petersburg dictionary measures the first process; completion scenarios and comparisons with the OED and the Chicago Assyrian Dictionary measure the second. Under PD's initial plan, completion falls around 2284; holding its present density across the alphabet moves the scenario to 3211. The paper then asks what digital resources can supply. DCS holds **25.2 %** of PD's primary citation mass, while **77.9 %** of DCS's 5.69 M tokens occur in the 118 PD-cited works it holds. An NCC commentary graph with 10,332 strict edges over 29,066 ranked works distinguishes commentary importance from citation mass and work extent. These evidence layers are joined in a five-table model linking works, editions, and resources. PD's central problem is not a shortage of quotations. It is the cost of converting an exceptionally rich apparatus into bounded, finished articles. Corpus coverage can relieve that cost; it cannot replace historical lexicography.

Keywords: Poona Dictionary, finishability, historical lexicography, citation density, source coverage, New Catalogus Catalogorum, Digital Corpus of Sanskrit.

1 Introduction

The entry *akāmataḥ* occupies three lines in the Petersburg dictionary and forty-seven in the Poona Dictionary (PD). Reviewing the latter in 1978, J. C. Wright asked whether such a twelve-fold increase in bulk could be welcomed wholeheartedly (Wright, 1978). The increase is real, but it is not uniform. PD carries 31 quotations and 75 attestations in that entry against two and six in the Petersburg dictionary. Across the lemmas shared by the two dictionaries, PD has 8.5 times the attestations in aggregate and a median of four times per lemma; one shared lemma in twelve falls below parity. Wright's bulk is, in effect, a citation apparatus. That apparatus can be enumerated, compared, and used to ask what kind of dictionary PD is becoming.

PD is *An Encyclopaedic Dictionary of Sanskrit on Historical Principles*, begun at Deccan College, Pune, in 1948 under S. M. Katre (Deccan College Department of Sanskrit and Lexicography, 2026; Katre, 1976). Its editors assembled ten million quotation slips by 1973, approximately ninety-five for every entry the dictionary would later print. The method had already been debated in Pune. A seminar held on 20–23 December 1972 produced "a detailed plan, a kind of blue-print, for a historical dictionary of Sanskrit" (Ghatage et al., 1973).[1] Yet the printed fascicles through *abhigamyamānaparadārādi* in 2019 remain within letter *a*. They have not passed *abhyāhita*, the last entry of Goldstücker's abandoned remake of Wilson.

[1] I bought my copy of this volume in Pune and plan to digitise it; it prints the sample articles examined by the participants.

This paper argues that PD's finishability depends on two linked processes. First, its lemmatisation and citation policies determine the amount of editorial work required per entry. Second, its editorial system must convert quotation slips into bounded, published articles. The initial thirty-seven-volume plan and the density actually printed imply very different outcomes: completion around 2284 under the plan, or around 3211 when current density is held across the remaining alphabet. These are scenarios, not date prophecies.

The same apparatus also makes a second question possible. PD supports its senses with quotations whose sigla identify cited works. The printed Index and Abbreviations of Books form a declared source bibliography; the works actually cited in the machine-readable letter-*a* text form a cited-source inventory. This inventory records editorial use, not a normative definition of Sanskrit literature. It is one more historical record of what a cosmopolitan literary culture used (Pollock, 2006). Comparing it with paper and digital resources shows which parts of that record can support lexical work today.

Three questions organise the study:

1. **What makes PD's entries so demanding to produce?** I compare PD and the completed Petersburg dictionary on shared lemmas and on a common quotation unit.

2. **What completion horizons follow from explicit design and production scenarios?** I contrast plan-based and density-held arithmetic, then test the OED and Chicago Assyrian Dictionary as comparative cases.

3. **How much of PD's cited-source base is available, and in what form?** I join PD to DCS and to the ACC/NCC bibliographic tradition, distinguishing citation mass, NCC commentary importance, work extent, and resource coverage.

The coverage measurements have a narrower scope than the finishability argument. They use the Cologne PDScan 2020 digitisation, whose fascicles issued 1976–2009 reach approximately *apaca-* and contain ∼104,959 lemmas. The print record had reached Vol. 35 and 6,056 pages by 2019, still within letter *a*. About 4,330 printed pages are represented in the digital text; some 1,700 are not. Every PD-side coverage number therefore concerns the cited-source inventory exercised in that text, never PD's published whole or its whole-alphabet practice (§7).

2 Related work

Historical lexicography supplies the paper's first axis: how entry design affects the possibility of completion. Wright's review of PD made the consequences of an expanding apparatus visible at entry level (Wright, 1978). Mayrhofer addressed the same design problem for a "large-corpus language" and completed KEWA and EWA within a single scholar's working life (Mayrhofer, 1956 1980; Mayrhofer, 1992 2001; Mayrhofer, 1980). Work on the OED provides the closest institutional comparison. Hoffmann evaluated its quotation base as a linguistic corpus (Hoffmann, 2004), while Brewer reconstructed the editorial decisions that produced it (Brewer, 2007). These studies treat the apparatus as a sample of language; the present paper also treats it as a measurable editorial workload.

The second axis is the evidentiary use of citation inventories. Corpus representativeness is usually prospective: designers balance registers, genres, and periods against a target population (Biber, 1993; McEnery and Hardie, 2012). Historical languages often lack that population. The Thesaurus Linguae Graecae addressed it through an explicit Canon of Greek Authors and Works (Berkowitz and Squitier, 1990), and Diorisis through documented literary-historical selection (Vatri and McGillivray, 2018).

Other research builds corpora from historical citation archives or links those archives to full texts (De Smet, 2005; Wills et al., 2018; Biffi et al., 2026). Within Sanskrit, prior work has measured citation registers inside digital dictionaries (sanskrit-lexicon project, 2026) and aligned dictionary sense inventories (Patel and Kulkarni, 2024).

DCS supplies the most advanced analysed corpus in this comparison. Its annotations support treebanks, neural sandhi segmentation, and lexical-semantic resources (Hellwig et al., 2020; Hellwig and Nehrdich, 2018; Hellwig and Biagetti, 2025). Its own documentation is equally important for interpreting coverage: Hellwig states that the selection reflects his research interests, is neither exhaustive nor representative of the whole body of Sanskrit literature, and contains incomplete texts (Hellwig, 2010). Two creator-side analyses address coverage at different units. Within the Vedic subcorpus, Hellwig, Nehrdich and Sellmer estimate that DCS's lexical and morpho-syntactic annotation covers about one quarter of the extant Vedic corpus, and show that DCS and the Vedic Treebank reflect scholarly preferences rather than balanced samples (Hellwig et al., 2023). At the semantic-annotation layer, Hellwig and Biagetti document period- and subject-specific coverage shaped by project-specific research priorities (Hellwig and Biagetti, 2025). Kulikov's handbook chapter supplies the wider survey of indigenous and digital Sanskrit lexicography (Kulikov, 2022).

The contribution joins these axes rather than presenting corpus coverage as an isolated result. It asks how PD's design and editorial history condition its finishability, then uses bibliographic and corpus evidence to assess what a corpus-welded lexical layer could change. Within a documented search of the ACL Anthology, arXiv, eLex, EURALEX, and ISCLS, I found no prior Sanskrit study joining those two questions.[2]

3 Data

PD side. The machine-readable PD text is the Cologne PDScan 2020 edition.[3] It covers the fascicles issued 1976–2009, from letter *a-* to approximately *apaca-*, with ∼104,959 lemmas. All PD-side numbers in this paper therefore concern **PD's citation-source inventory as exercised under the letter *a-* in that text**, not its full declared canon (§7).

DCS side. The analysis uses the bounded inventory of 276 texts in the delta table, 270 of them token-bearing. Token counts come from the per-text token-delta table committed in the VisualDCS repository at two snapshots: DCS 2021 (4.58 M tokens) and DCS 2026 (5.69 M tokens).[4] Chapter counts come from the DCS 2021 metadata and the DCS abbreviation list.

Bibliographic graph. The NCC layer measures commentary importance, not resource availability. Its strict commentary graph contains 10,332 deduplicated edges over 29,066 ranked works; a fuzzy variant supplies a recall ceiling. Raw commentary-tree size and PageRank remain separate measures. ACC and NCC provide bibliographic witnesses, while PD and PWG provide independent lexicographic citation inventories.

Five-table evidence model. The analysis links five tables: *Works*; *Commentary relations*; *Editions/Witnesses*; *Resource observations*; and *Resources*. NCC importance attaches to the abstract work. Extent and completeness attach to an edition or witness. Availability attaches to a resource observation. The main chain is therefore Work → Edition/Witness → Resource observation → Resource; every figure is derived from these linked tables rather than maintained as an independent source of truth.

Committed intermediate data. All derived tables are committed in the public `csl-atlas` repository under `data/pd/`. These include the raw siglum census (`pd_siglum_raw.tsv`: 5,106 distinct sigla, 398,359 occurrences) and the classified census (`pd_siglum_families.tsv`). The DCS join (`pd_dcs_text_crosswalk.tsv`) records 118 covered texts with per-snapshot token counts and a coverage grade; the headline metrics are in `pd_dcs_metrics.json`. Extraction and join scripts (`pd_extract_sigla.py`, `pd_dcs_crosswalk.py`) are committed alongside.

[2] The search log and neighbour table are available from the author on request. The Semantic Scholar full index returned HTTP 429, and Kulikov's chapter was checked at abstract and bibliography level because its full text was inaccessible on 26 October 2026. These access limits preclude an absolute field-first claim.

[3] `https://sanskrit-lexicon.uni-koeln.de/scans/PDScan/2020/web/index.php`; 55 MB, 107,630 `<L>` entries; © 2014 The Sanskrit Library and Thomas Malten, CC BY-NC-SA 3.0; read-only external source.

[4] `https://github.com/gasyoun/VisualDCS/blob/main/derived-data/Corpus-Delta-2021-2026/per_text_token_delta.csv`

Siglum adjudication. Siglum families are adjudicated against PD's own abbreviation list (the declared-canon join of §7); near-homographs remain separate rather than being merged by surface similarity. The sharpest case distinguishes *MahāBhā.* (some nine thousand occurrences), the DCS-covered *Mahābhārata*, from *MahāBh.* (under two thousand), Patañjali's *Mahābhāṣya* in the same abbreviation list. The epic is not declared under *MahāBh.*, so the pair is never merged and *MahāBh.* remains residue. The diacritic-loss variant *MahaBhā.* (12 occurrences, title unresolved) likewise remains unmerged and in the residue. The siglum *Bharat.* (3 occurrences) also remains in the residue, but as a distinct work rather than a rival epic name. The same abbreviation list declares it the *Bharatakadvātriṃśikā* (Messrs A. M. & Co., Palitana), cited by folio and line. Its three locators, *1B.* 6, 11. 11, and *9A.* 6, are two-component and folio-suffixed, not the epic's *parva.adhyāya.śloka.* The *Mahābhārata*, by contrast, is declared under *MahāBhā.* (BORI 1927–1961; the same nine thousand occurrences, DCS-covered), so no epic occurrence hides under *Bharat.* One qualification follows for the headline claim of §4.2: although 100 % of occurrence mass carries a *class*, unresolved *titles* are not thereby confirmed.

4 Method

4.1 Citation-source inventory and declared-canon criteria

The study uses two registers deliberately. **Declared canon** names only the works printed in PD's Index of Works or Abbreviations of Books. The exercised letter-*a* apparatus is a **citation-source inventory**, used as a canon proxy; it is not a second editorial declaration. Its membership criteria are stated here explicitly:

> **PD's exercised citation-source inventory**, as operationalised in this paper, is **the set of Sanskrit works that the machine-readable letter-*a*- text cites as primary sources**, weighted by citation occurrence count. A citation-siglum occurrence enters the inventory denominator **iff** it is adjudicated class `primary` under the three-way scheme of §4.2. A *work* is a member of the inventory iff at least one `primary` occurrence resolves to it.

Three consequences of this definition are intended and worth surfacing:

1. **The inventory is frequency-weighted by the dictionary's own practice.** A work PD cites 3,506 times (Padmapurāṇa) counts 3,506× in the citation-weighted metric. This is frequency salience in documented practice, not an editorial canonicity ranking imposed here.

2. **Books *in* Sanskrit, not books *on* Sanskrit.** Modern scholarship PD cites (etymological dictionaries, epigraphic corpora, concordances) is excluded from the inventory as class `secondary` (§4.2). The inventory is the object-language literature.

3. **The inventory is scoped to the digitised fascicles.** PD under letter *a*- exercises ∼2,445 distinct works (skeleton-merged estimate; §5.2); the printed declared lists are larger. I measure the exercised inventory, which is what the data supports; the scope caveat is carried explicitly throughout (§7).

4.2 Classification of siglum occurrences

Every one of the 398,359 siglum occurrences (Figure 1 exhibits them in situ, in the openings of two of the dictionary's largest citation-bearing entries) is assigned exactly one class:

A siglum occurrence is not the same thing as a quotation, and the difference is measured. A quotation is a maximal run of consecutive SLP1 blocks separated only by whitespace and closed by a siglum-plus-locus. The block itself is not the unit, because it also carries the headword, the morphemes of an etymology, and the two halves of one passage broken at a line

Class	Criterion	Examples
primary	the siglum names a Sanskrit work (any period, any genre, including commentaries and śāstra)	*MahāBhā.*, *ṚV.*, *PadmP.*, *Kāśi.*
structural	the token is citation apparatus, not a work: grammatical case labels, locus sub-parts, Roman-numeral book/volume numbers, editorial markers	*Acc.*, *A.*, *iii.*, *Ed.*, *App.*
secondary	the siglum names modern scholarship *about* Sanskrit	*EI.* (Epigraphia Indica), *MW.*, *PW.*, *POK.*, *VIŚVA.*

wrap (the largest entry, *anya*, holds 3,904 blocks). On the same text, the letter-*a*- citation-source inventory holds 302,442 quotations and 419,890 attestations across 107,630 entries. The attestations comprise 384,217 siglum-bearing loci plus 35,673 further references that inherit a siglum after a semicolon (Reader's Guide X.v). The text averages 2.05 blocks per quotation; 1,686 entries carry no quotation at all. Of the 398,359 siglum occurrences classified here, 384,217 close a quotation and the remainder attach to a bare reference or to an entry without one. The resulting 96.5 % agreement between two independent tokenisations of the same text is the cross-check for this classification. Unit definition and the per-entry table: `https://github.com/sanskrit-lexicon/csl-atlas/tree/main/data/pd`.

Primary occurrences are further split by the DCS join into **covered** (the work exists in DCS's 276-text inventory) and **residue** (it does not). Every occurrence carries a class, so the classification reaches 100 % adjudication of occurrence mass and clears the ≥95 % stop-condition set in the study design.

4.3 Design: anchor on the bounded side

PD's citation mass has a very long tail: the top 300 sigla carry only ∼74 % of it, so expanding every siglum to a title (5,106 sigla, most rare, many abbreviation variants) is infeasible and unnecessary. The join is therefore anchored on **DCS**, which is bounded: each of the 276 DCS texts is mapped to its PD siglum(s), making the *covered* mass exact at any frequency rank. Everything primary that no DCS text claims falls out as the residue. The cost of this design is asymmetric confidence: covered-set identifications are individually hand-identified, while the residue's *title-level denominator* (how many distinct works the residue represents) must be estimated from siglum-variant merging (§5.2). I accept that cost because both headline metrics (citation-weighted, token-weighted) depend only on the exact covered mass.

4.4 Metrics

1. **PD-citation-weighted coverage** = covered primary occurrences / all primary occurrences. *Of what PD cites, how much is in DCS?*

2. **Title-level coverage** = covered distinct works / distinct works PD cites (lower bound over raw sigla; estimate over consonant-skeleton-merged sigla). *How many of PD's works exist in DCS at all?*

3. **DCS-token-weighted coverage** = tokens in PD-cited DCS texts / all DCS tokens, at each snapshot. *Of DCS's own mass, how much does PD also cite?*

4. **Partial-coverage grade**: per covered text, whether DCS holds the work or a fragment, sized by `dcs_tok_2026` against the work's known extent.

```
agni
{#agni#}¦ (agní) {%m.%} [IE. MAYR. DEBRU. p. 741; TURN. 55; Pok. 293 /
{#tadvA\ enametada\gre devA\nAmajanayata.ta\smAdagni\ragri\rha vE nA\mEtadya\dagniri\ti.sa\ jAtaH#}
{#pU\rvaH pre\yAya yo vE pU\rva etya\gra etI\ti vE ta\mAhuH\ soe\\vAsyAgni\tA#} ŚatBr. II. ii.
4. 2; {#a\Ta yo ga\rBo'ntarAsIt.so'gri\rasfjyata sa ya\dasya sa\rvasyA\grama\sfjyata ta\smAda-#}
{#gri\ragri\rha vE ta\magnirityA\cakzate paro\'kzam#} ŚatBr. VI. i. 1. 11; {#agniH kasmAt.agraRI-#}
{#rBavati.agraM yajYezu praRIyate.aNgaM nayati sannamamAnaH.aknopano BavatIti#}
{#sTOlAzWIviH.na knopayati na snehayati.triBya AKyAteByo jAyata iti SAkapURiH.#}
{#itAt.aktAddagDAdvA.nItAt.sa KalveterakAramAdatte gakAramanaktervA dahatervA nIH#}
{#paraH#} Nir. 7. 14 (139. 6); BṛDe. 2. 24; aṅg+ni, Uṇā. 4. 50; SiddhāKau.
550B. 3; ({#aNgatyUrDvaM yAtItyagniH#} AmaKoUd. 17. 14 ({%on%} i. 1. 53)); {#agniritya-#}
[... the agni entry continues for another ~139,000 machine characters]

ap
{#ap#}¦ (ap) {%f.%} [takes {%‘yat’%} suffix in the sense of purified, P. iv. 4. 134; affix
{%‘a’%} comes after {%‘ap’%} and becomes {%‘apa’%} in the final of a compound, P. v. 4. 74;
JaineVyā. iv. 2. 70; CāndraVyā. iv. 4. 57; ŚākaṭāVyā. ii. 1. 39; SaraKaṇṭhā.
(Gr.) v. 4. 86; SiddhaHe. vii. 3. 76; MugdhaBo. 6. 69 (62. 12); with case endings
takes acute accent, P. vi. 1. 171; KāśiVṛ. {%on%} vi. 1. 171; long {%‘ī’%} is the substitute
for {%‘a’%} in {%‘ap’%} and the form is {%‘īpa’%} coming after {%‘dvi’, ‘antar’%} and
prepositions, P. vi. 3. 97; CāndraVyā. v. 2. 113; ŚākaṭāVyā. ii. 2. 138; SiddhaHe.
[... the ap entry continues for another ~151,000 machine characters]
```

Figure 1: The unit of analysis, exhibited. Verbatim openings of the machine-readable PD entries *agni-* (p. 1-0329a) and *ap-* (p. 8-4266b), two of the dictionary's largest (140k and 152k machine characters; both cut here). The markup is the digitisation's own: brace-hash-delimited spans carry Devanāgarī-encoded text, brace-percent-delimited spans are italics; the romanised citation sigla in situ (*ŚatBr.*, *Nir.*, *P.*, *CāndraVyā.*, …) are the tokens the alignment counts. Source: PDScan 2020 (© The Sanskrit Library / Thomas Malten, CC BY-NC-SA 3.0), entry ids 9424 and 106685.

4.5 Reproducibility

All four figures and every number in §5 regenerate from the committed data via `python scripts/build_a68_figures.py` in the `csl-atlas` repository. The script re-derives the metrics from the row-level TSVs and cross-checks 12 values against the committed `pd_dcs_metrics.json`. It aborts on any mismatch, so the paper's figures cannot silently drift from the data.

5 Results

5.1 The design workload: density before coverage

PD's first letter contains 104,959 lemmas, approximately 99 % of the completed PWG's headword count, but only 4.3 % of PWG's alphabetical ground. On the shared span it is 23.2× as dense. This is a difference of editorial policy, not a simple claim that PD contains more evidence overall. PWG still carries more attestations across the whole alphabet: PD's letter-*a* apparatus holds 419,890 attestations against PWG's 729,719 quotation-linked attestations. PD's distinguishing feature is the concentration of work at each headword.

Three lemmatisation families show that policy in operation (Appendix B). PD heads a lemma for nearly every compound and productive derivative, while PWG selects a subset. The privative *a-* and negative *an-* account for about one PD headword in eleven. Removing them reduces the density multiple only from twenty-three to twenty-one. The design workload therefore survives the most obvious alphabetical confound.

The same-unit comparison also disciplines Wright's example. At *akāmataḥ*, PD has 12.5 times the attestations, but the median shared lemma has four times, and one shared lemma in twelve has fewer. PD's exceptional density coexists with a long tail of thin compound entries. Any completion model must therefore reckon with both the heavy entries and the cost of admitting so many light ones.

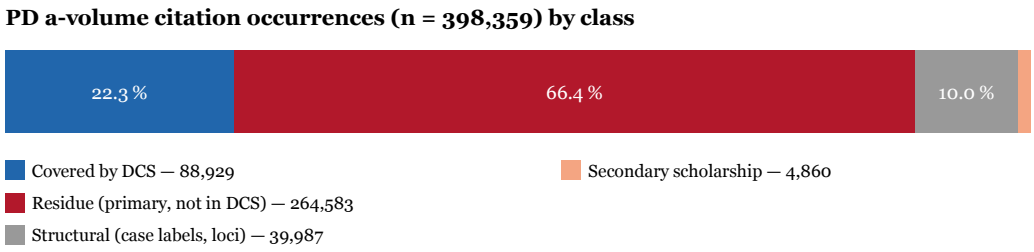


Figure 2: Mass breakdown of the 398,359 citation-siglum occurrences in PD's letter-*a*- volumes: primary (citation-source denominator), split into covered and residue, against structural (apparatus) and secondary (modern scholarship) tokens; the covered slice is 22.3 % of all occurrences and 25.2 % of the primary ones. Scope: all quantities are measured on the machine-readable PD text (fascicles issued 1976–2009, letter *a*- to *apaca*-); the print record (Vol. 35, 6,056 pp by 2019) is not included.

The initial thirty-seven-volume plan and the observed density yield two completion scenarios. The plan points to approximately 2284. Holding present density across the remaining alphabet points to approximately 3211. These scenarios separate the size of the design commitment from the annual production rate: scope changes the horizon far more than a plausible improvement in pages produced per year. The comparison with PWG's twenty years to completion is thus not a moral contrast between industrious and stalled scholars. It is a measurement of what different editorial units make finishable.

5.2 Mass breakdown

Of 398,359 siglum occurrences in PD's letter-*a*- volumes (Figure 2):

Class	Occurrences	Share of all
primary: the citation-source denominator	353,512	88.7 %
of which covered by DCS	88,929	22.3 %
of which residue (not in DCS)	264,583	66.4 %
structural (apparatus tokens)	39,987	10.0 %
secondary (modern scholarship)	4,860	1.2 %

The "books *in* Sanskrit, not books *on* Sanskrit" framing survives contact with the data: secondary scholarship is a real but small (1.2 %) minority, now quantified and excluded from the citation-source inventory.

5.3 The three coverage metrics

Figure 3 shows the headline result:

#	Metric	Value
1	PD-citation-weighted coverage	**25.2 %**
2	Observed title-level overlap	**8.0 %** lower bound (118 of ≈1,476 AOB entry rows; band 7.8–8.3 %); exercised letter-*a* band **2.4–4.8 %** (118 of ∼2,445)
3	DCS-token-weighted coverage	**77.9 %** (2026) · 74.1 % (2021)

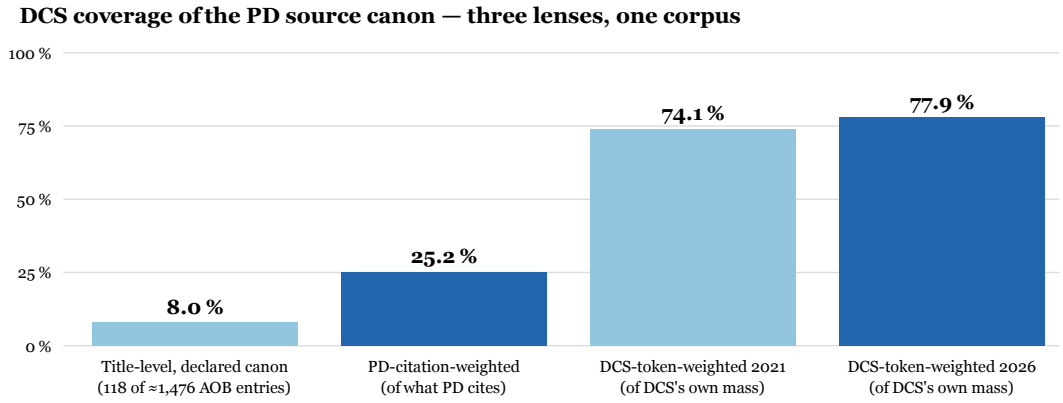


Figure 3: DCS coverage of PD's citation-source inventory under three metrics: citation-weighted (25.2 %), observed title-level overlap (an 8.0 % lower bound, band 7.8–8.3 %; exercised letter-*a* band 2.4–4.8 %), and DCS-token-weighted (77.9 % in 2026, 74.1 % in 2021). The four bars do not share a denominator, and are not meant to. Scope: all quantities are measured on the machine-readable PD text (fascicles issued 1976–2009, letter *a-* to *apaca-*); the print record (Vol. 35, 6,056 pp by 2019) is not included.

Metric 1. PD's primary citation occurrences point at works DCS contains in 25.2 % of cases. This is any-presence coverage: a work counts as covered even when DCS holds only a fragment. Excluding the two explicitly partial texts, Skandapurāṇa and Kāśikāvṛtti, gives a strict sensitivity value of 23.3 %. The two measures therefore answer different questions, and 25.2 % must not be read as complete-work coverage.

Metric 2. DCS contains 118 of the distinct works exercised in PD's letter-*a* text. These works supply an observed lower bound of **8.0 %** on overlap with the approximately 1,476 AOB entry rows, not a complete measurement of DCS coverage of that bibliography. The full declared list has not undergone the same DCS join. Against the cited-source inventory exercised in the letter, the corresponding band is 2.4–4.8 %. The AOB, Index of Works, and exercised sigla are different populations, shown separately in Table 1. The clean-entry rows of `pd_aob_declared_works.tsv` are a floor, never a denominator, and the Index recomposition remains posed rather than run (Appendix A).

Base	Rows	Status
AOB central estimate (the paper's denominator)	≈1,476	the quoted denominator; band 7.8–8.3 %
AOB entry rows (row test, header rows removed)	1,469	118 / 1,469 = 8.03 %, displayed as 8.0
AOB audit bracket against the printed pages	1,419–1,517	the ends of the band
AOB clean-entry filter	1,112	a floor, never a denominator (10.6 % would overstate)
Index of Works, all printed works	2,031	work authority; recomposition not run (109 of the 118 join by exact siglum)
Index of Works, rows with a siglum	1,622	as above
Exercised letter-*a* sigla, raw	4,988	over-counts works: 118 / 4,988 = 2.4 %
Exercised letter-*a* sigla, skeleton-merged	∼2,445	estimate: 118 / 2,445 = 4.8 %

Table 1: The bases of the declared and exercised denominators, and what each supports. The bases are not nested: the printed lists count works, the exercised count counts surface siglum families and over-counts works, so the ratios are not bounds on one another.

Metric 3. Weighting by DCS's own token mass reverses the perspective: **77.9 %** of its 5.69 M

tokens, or 4.43 M, lie in PD-cited texts. The Mahābhārata alone supplies a fifth of the corpus. Thus DCS's token mass overlaps heavily with works PD also cites, even though most of PD's broader citation practice falls outside DCS. This is composition evidence, not proof that DCS represents Sanskrit literature as a whole.

Metric 4. Most covered texts are graded `present`; Skandapurāṇa and Kāśikāvṛtti carry explicit partial flags. The crosswalk records per-text token and chapter counts so that presence can be assessed against extent. The 23.3 % sensitivity value above removes these two partial witnesses rather than allowing a fragment to stand for a complete work.

Two denominators must remain distinct. The declared source bibliography lists works in print; the letter-*a* citation mass records what the machine-readable text exercises. A ratio may not take its numerator from one population and its denominator from the other. Appendix A therefore reports the ACC/NCC composition as interim and does not treat it as a coverage percentage.

5.4 The residue: what DCS structurally misses

The most informative output is the ranked list of works PD cites most that DCS lacks (Figure 4; full data in `pd_siglum_families.tsv`). The top of the list is the Padmapurāṇa, then the Mahābhāṣya, the Brahmāṇḍapurāṇa, the Rājataraṅgiṇī and the Bhaviṣyapurāṇa, with the Vaijayantī, the Śiśupālavadha, the Kādambarī and the Raghuvaṃśa behind them (Figure 4 ranks them by PD's own count). Four clusters account for nearly all high-frequency residue and identify its main structural gaps relative to a historical dictionary's citation-source inventory:

1. **Purāṇas.** DCS holds ∼12 purāṇas; PD cites at least a dozen more it lacks (Padma, Brahmāṇḍa, Bhaviṣya, Brahma, Mārkaṇḍeya, Vāyu, Gaṇeśa, Viṣṇudharmottara …). The single largest residue item, the Padmapurāṇa, outweighs all but three *covered* texts.

2. **The indigenous lexicographic tradition (kośa).** Vaijayantī, Medinī, Nānārthasaṃgraha, Anekārthasaṃgraha, Viśvaprakāśa: a dictionary naturally cites other dictionaries; a corpus of running text holds almost none.

3. **Classical kāvya and nāṭaka.** No Raghuvaṃśa, no Kādambarī, no Śiśupālavadha, no Naiṣadhīyacarita, no Mṛcchakaṭika, no Mudrārākṣasa: the mahākāvya/drama backbone of classical literature is largely outside DCS.

4. **The grammatical/śāstric commentary layer.** The Mahābhāṣya, the Vārttikas, and the dense bhāṣya/ṭīkā apparatus a historical-principles dictionary leans on.

DCS returns nothing for the word *atiyuṣmad*. This is neither a rare word misspelled nor a form the lemmatiser failed on: the string does not occur in the corpus at all, nor does any sandhi variant of it, and the lemma is absent from its inventory. PD, meanwhile, gives the word 123 printed lines and 109 quotations. The entry is a grammatical construct: *ati* compounded with the pronoun *yuṣmad*, generated so that the description can decline a compound governed by its first member. The page supplies the paradigm, singular, dual, and plural of the subordinate referent through every case. Its citations are the rules that generate the forms, the Mahābhāṣya's own paradigms first, then the Kāśikāvṛtti, the Nyāsa, the Jaina and Śākaṭāyana grammars, the Siddhāntakaumudī and the Prakriyākaumudī, with two Vedic *cf.* citations for the pronoun bases. Here is the asymmetry the residue is made of, at the scale of one word. PD documents texts and the apparatus that makes them describable; DCS holds the former but not the latter in this case. The residue is therefore generic as well as bibliographic: the grammatical commentary layer is absent as a genre, not merely as a list of titles.

The residue is therefore a candidate acquisition queue, ranked by the gain each addition would make to PD-citation-weighted coverage. It is not a universal ranking of scholarly importance or digitisation feasibility. NCC importance, work extent, available editions, and annotation cost must be assessed before assigning priority (§6.3).

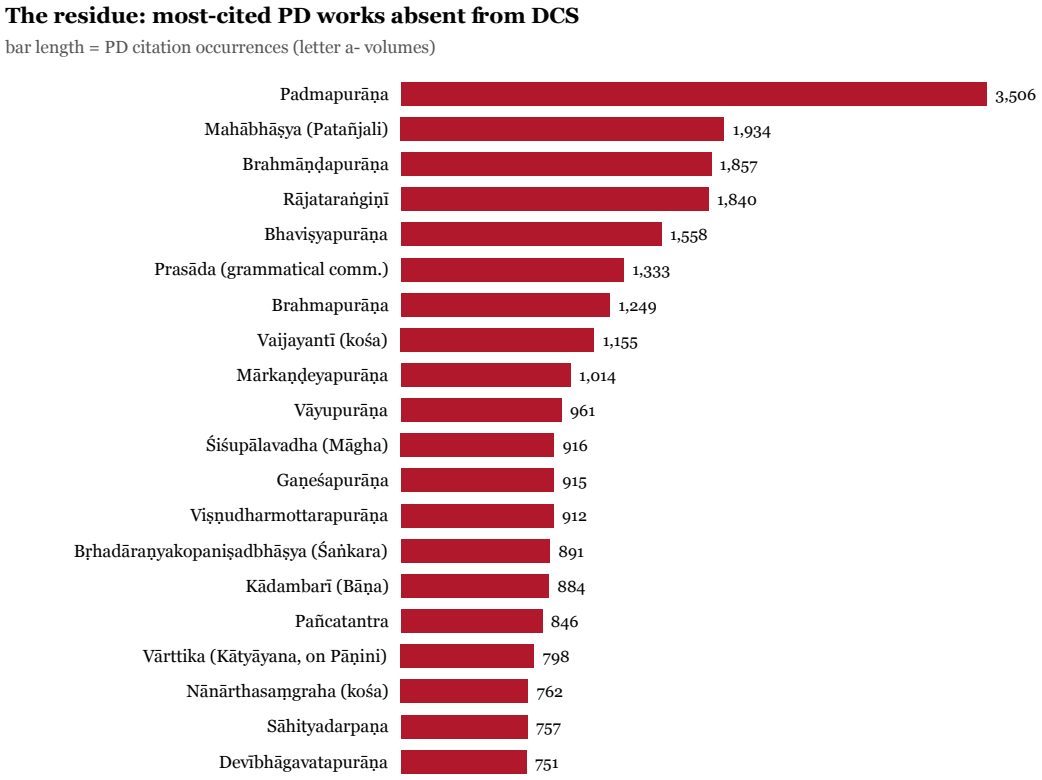


Figure 4: Top-20 residue: the works PD cites most that DCS lacks, ranked by PD citation frequency. The ranking supplies citation-mass input to an acquisition queue, not a complete priority ranking. Scope: all quantities are measured on the machine-readable PD text (fascicles issued 1976–2009, letter *a-* to *apaca-*); the print record (Vol. 35, 6,056 pp by 2019) is not included.

5.5 NCC importance and four resource witnesses

The NCC graph supplies an independent importance axis. The Amarakośa leads its strict commentary-tree ranking, while the Mahābhārata leads PD's citation-mass ranking. Commentary importance and lexicographic attention are therefore not interchangeable. Table 2 shows the share of the top-N strict commentary-ranked works present in four witness sets. These are title-level overlaps, not occurrence-weighted metrics and not a league table of resource quality.

Witness	Top 50	Top 100	Top 250	Top 500	Top 1000
DCS	16.0 %	13.0 %	9.2 %	5.6 %	4.0 %
PD (adjudicated titles)	20.0 %	13.0 %	10.4 %	7.2 %	5.0 %
PWG citation inventory	32.0 %	27.0 %	19.6 %	13.2 %	9.1 %
ACC manuscript catalogue	86.0 %	86.0 %	78.4 %	74.2 %	67.7 %

Table 2: Coverage of the strict NCC commentary-importance ranking by four witness sets. ACC's manuscript-catalogue coverage remains high, while DCS, PD, and PWG decline as the importance ranking expands. Each join has its own scope and denominator; PD remains letter-*a*-only and PWG remains full-alphabet.

ACC's stability shows that the commentary canon remains well represented in the manuscript catalogue tradition. The lower and falling DCS, PD, and PWG coverage indicates a different selection pressure. Analysed corpora and lexicographic citation inventories favour the Vedic, epic, and ritual core more strongly than the lexico-grammatical and liturgical works that generated large commentary traditions. The graph thus identifies a gap that DCS-only coverage cannot describe.

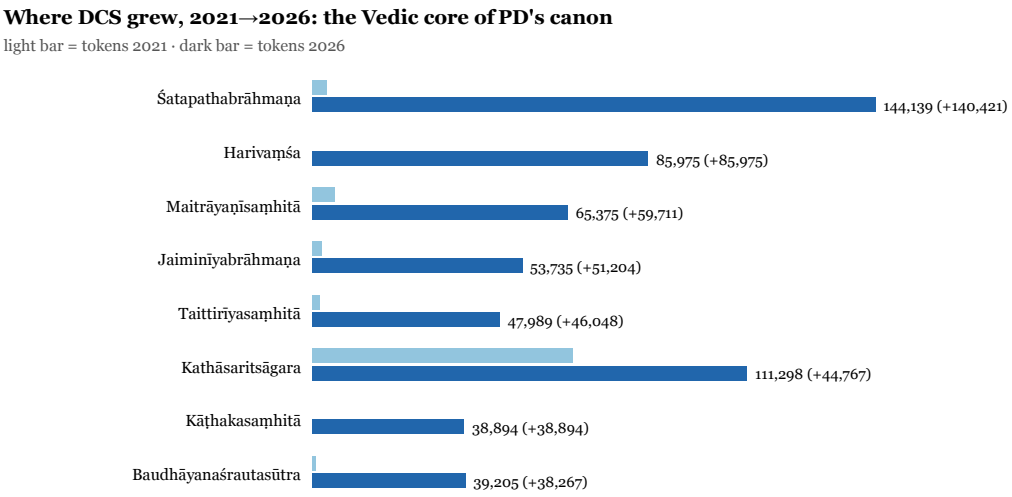


Figure 5: DCS growth 2021→2026 by text: the added token mass concentrates in Vedic prose and epic, both PD-core works, so token-weighted coverage of PD's citation-source inventory rises 74.1 %→77.9 %. Scope: all quantities are measured on the machine-readable PD text (fascicles issued 1976–2009, letter *a-* to *apaca-*); the print record (Vol. 35, 6,056 pp by 2019) is not included.

5.6 The 2021→2026 delta: the gap closes from the Vedic end

Token-weighted coverage rose from 74.1 % to 77.9 % (+3.8 pp) across the two snapshots because DCS's growth was concentrated in PD-core texts, overwhelmingly Vedic prose (Figure 5). The like-for-like 2021 denominator excludes four commentaries and two zero-token stubs dropped by DCS 2026. The raw 2021 baseline is 74.0 % and yields the same rounded delta (`https://github.com/sanskrit-lexicon/csl-atlas/blob/main/data/pd/pd_dcs_2021_baseline.json`). The Śatapathabrāhmaṇa grew from under four thousand tokens to a hundred and forty-four thousand, and the Harivaṃśa from nothing to eighty-six thousand. Covered-text tokens grew from 3.39 M to 4.43 M in five years.

Citation-weighted coverage moved from 21.9 % in 2021 to 25.2 % in 2026, a +3.3 pp change. Two snapshots establish the direction of change, not a growth law. The large additions also show why linear extrapolation would be misleading: one digitisation can move the metric discontinuously.

6 Discussion

6.1 Coverage, importance, and complementarity

The two DCS metrics answer different questions. PD's citation practice spans an encyclopaedic inventory, so a corpus can overlap with its frequent core while missing most of its citation mass. Conversely, most of DCS's own tokens lie in works PD also cites. DCS is therefore aligned with part of PD's core, not with its breadth. For morphology, frequency norms, and the epic register this overlap is useful. For purāṇic, kośa, kāvya, and commentary-based historical semantics, the corpus must be supplemented. The 23.3 % strict sensitivity value marks where partial witnesses limit even that core alignment.

The kośa gap has a generic explanation as well as a numerical one. Kośas are versified, memorisable synonym dictionaries shaped by a manuscript ecology (Vogel, 1979). A corpus designed around continuous running text will not naturally contain them. PD, by contrast, cites them as lexical evidence. This is one reason a historical dictionary and a running-text corpus are complements rather than substitutes.

The declared source bibliography has its own shape. PD's printed *Index of Works*

(pp. LXXXIX–CXXVII), transcribed for this study, classifies 2,001 of its 2,031 works by subject (Table 3).

Class (Index of Works)	Works	Class (Index of Works)	Works
Minor Upaniṣads	175	Kāvyas	75
Vedānta (all schools)	173	Śrauta and Gṛhya sūtras	69
Jainism	97	Other poems	66
Stotras, stavas	91	Tantras	63
Dramas	85	Lexicons	56
Anukramaṇīs	77	Dharma	55
Poetics	77	Buddhism	52
Grammar (Pāṇinian)	76	Purāṇas	44

Table 3: The declared canon by subject, from PD's own *Index of Works*: 2,001 of 2,031 printed works carry a subject class, and the sixteen largest classes are shown. The classes are the catalogue's, not this paper's; the counts are the transcription's, re-derived from `pd_index_works.tsv`.

The bibliography is devotional, scholastic, and literary. Upaniṣad and Vedānta classes contain 348 works; kāvya, drama, poetics, stotra, and miscellaneous poetry contain 394; Jain, Buddhist, and tantric classes contain 212. Śrauta and Gṛhya sūtras account for 69, or 3.4 % of the classified list. The bibliography therefore includes substantial layers that a Vedic-epic corpus does not prioritise. Its dictionary-wide composition should not be confused with the letter-*a* measurement, whose privative and preverb vocabulary is also non-random.

6.2 Design workload and editorial conversion

Finishability also has an editorial history. In 2003, *Outlook* reported both a five-year demand for all volumes and a "good 50 years" forecast. Its own figure of 16,632 pages left at 500 pages a year implies 33 years, so both remain attributed press claims rather than realized rates (Kabra, 2003). The author's 2019 operational account records a 200-print-page annual target under Prasad Joshi, general editor since 2017; that target has not been realized either (Gasūns, 2026). These documents justify scenario analysis, not a prophecy.

The comparative record shows why the entry unit matters. Grimm took 123 years, WNT 134, TLL remains unfinished after 156, and the OED took 71. PD's density-held scenario exceeds all of them because its entry design requires more decisions per headword. Mayrhofer's monograph on designing a dictionary for a "large-corpus language" names the underlying problem (Mayrhofer, 1980). KEWA and EWA show that a bounded design can be completed by one scholar in 24 and 15 years respectively (Mayrhofer, 1956 1980; Mayrhofer, 1992 2001).

The OED supplies the institutional comparison. Conceived in 1857, it spent twenty-two years with more than eight hundred volunteer readers and no dictionary to show for them. It lacked a bounded plan, an editor empowered to absorb the slips, and a publisher's deadline. The eventual system made contribution mechanical: assign a book, copy a quotation, file a headword, and give one editor authority to consume the result. PD completed the first half of that system at extraordinary scale. A slip is a promise; a dictionary is a decision. The comparison suggests that PD's bottleneck lies not in the supply of quotations but in the mechanism for converting them into articles.

PD's density against the finished PWG, and the arithmetic behind its claim to three times the OED, are set out in Appendix B. The claim survives as a volume count, not as a like-for-like measure of size.

The Chicago Assyrian Dictionary supplies the counter-example. It began in 1921 and finished in 2011: ninety years, twenty-one volumes, twenty-eight thousand words (`https://news.uchicago.edu/story/huge-dictionary-project-university-chicago-completed-after-90-years`). PD's letter *a* alone contains 104,959 lemmas, nearly four times that word list, while remaining under

five per cent of its alphabet. A long project can finish. The decisive difference is the unit of work and the institution's willingness to bound it.

Goldstücker makes the point genealogical as well as numerical. His remake of Wilson (1856–1864) ended at *abhyāhita* after some 6,800 exhaustive entries. PD's printed fascicles still have not passed that entry. Its 104,959 letter-*a* lemmas approximate the whole PWG and two-thirds of PWK, for under five per cent of the alphabet.

The corpus comparison does not imply that ingestion should replace lexicography. DCS adds texts quickly; PD deepens historical sense-development, etymology, and homonym adjudication. These are different forms of production. A corpus-welded lexical layer can reduce duplicated reading and make coverage explicit, while the dictionary remains responsible for analysis.

6.3 From residue ranking to acquisition policy

The Padmapurāṇa offers the largest single gain to PD-citation-weighted coverage, and the top-20 residue works together carry some twenty-four thousand citation occurrences. That makes them candidates for acquisition, not automatic priorities. A responsible policy must also consider NCC commentary importance, work extent, available editions, and the cost of producing analysed text.

The GRETIL join changes the task for part of the queue. Much of the top-20 residue is already machine-readable and requires ingestion rather than fresh digitisation. Six works remain absent from both DCS and the GRETIL snapshot of 14 October 2026: the Padmapurāṇa, Bhaviṣyapurāṇa, one grammatical commentary, Vaijayantī, Gaṇeśapurāṇa, and Nānārthasaṃgraha, with a little over nine thousand citations in all.[5] The task therefore divides into four classes: acquire, complete, align, and annotate.

Access is an institutional constraint as well. The Pune scriptorium's intranet means that PD's machine text reaches scholarship mainly through third-party digitisation. The *kūpamaṇḍūka* metaphor names that isolation, but the remedy is reciprocal: dictionary and corpus projects must make their holdings addressable to one another.

DCS now enters a domain-adapted translation model as an annotated CoNLL-U corpus (Nehrdich and Keutzer, 2026). The works missing from DCS are therefore absent from that model as analysed text, not merely as raw strings.[6] Digitisation alone does not close this gap. The text must also be annotated and published.

7 Limitations

- **Letter *a*- only, with 2009-vintage machine text.** PD was conceived by S. M. Katre in 1948; its scriptorium holds ∼10 million quotation slips; the official ledger documents 35 fascicle-volumes / 6,056 pages by 2019, all still under letter *a*- (through *abhigamyamānaparadārādi*; (Deccan College Department of Sanskrit and Lexicography, 2026; Deccan College Postgraduate and Research Institute, 2026)). The machine-readable text measured here (Cologne PDScan 2020) covers the fascicles issued 1976–2009, *a*- to ∼*apaca*- (∼105k lemmas). Every number here concerns PD's **citation-source inventory as exercised under letter *a*- in that text**, not its full declared canon or whole-alphabet practice. This is the single biggest caveat. The letter is not a random slice: privative and preverb compounds shape its citation contexts, and purāṇas, kośa, kāvya and commentary are missing wherever they fall alphabetically. The apparatus is also imperfect as a record: the Manu 11.127 attestation PWG carries at *akāmataḥ* and PD drops is absent from the machine text as well. Nothing in the letter-*a* measurement licenses whole-alphabet extrapolation.

- **The declared canon is itself an OCR-dependent object.** The printed front matter,

[5] Only the top-20 residue works were crossed against the GRETIL snapshot. Rows 21–44 remain uncrossed, so this is not a completeness claim or a commitment to a digitisation date.

[6] Nehrdich and Keutzer report 0.61 B tokens for this component. That figure counts serialised CoNLL-U annotations under a different tokenizer and is not comparable with the 5.69 M running-text tokens used here.

the "List of Works and Abbreviations", is in hand: OCR'd from the print and published as a PDF by Deccan College's KoshaSHRI portal (accessed 14-10-2026), with the publisher and series lists beside it. An official-PDF re-OCR supplies reference readings for the parse's known defects: the sigla *ṚV.* and *AV.*, the Viṣṇupurāṇa row, and the *NyāySā./NyāySū.* pair. The corrected declared canon is an **entry**-level count of ≈1,476 rows, the central estimate of a 1,419–1,517 bracket. No committed parser reproduces a point estimate, so the paper quotes the central value with its band. The machine-readable siglum→title join and the 1,112 clean siglum entries are published in csl-atlas `data/pd/`. The clean entries are a floor, not the full declared list. A machine text is never finished, so every number in this paper measures a *text*, not the dictionary.

- **Sourced siglum→title expansion, partially.** At measurement time no machine-readable siglum→title table existed, so identification for the covered set was performed by hand in a single annotation pass, anchored on DCS's bounded inventory. No independent second annotation pass was conducted. All 120 hand-identified covered sigla join a declared row; 14 are anchored directly on the official PDF. The declared list is now machine-readable, and the census joins it at 63.3 % of sigla / 78.0 % of occurrence mass. Joins beyond the covered set are machine-fuzzy (≥0.85 folded similarity), not verified: the residue's title-level identifications are sourced leads, not adjudicated facts.

- **The two-catalogue composition is provisional.** The interim table of Appendix A uses confirmed crosswalk rows only, measures the direct PD→NCC path at title level only, and predates the siglum join that the printed *Index of Works* makes possible. Its counts are reported as counts, never as coverage percentages, and every figure is flagged *interim, AOB denominator*.

- **Coverage ≠ locus alignment.** I measure whether DCS *holds* a work PD cites, not whether DCS's edition matches PD's cited edition line-for-line. Partial-coverage grades (§5.2, metric 4) bound but do not eliminate this gap. The C10 full sample (100 PD citations × 5 covered texts) quantifies both halves. In all, 74 of 100 sampled loci resolve in DCS; the misses sit exactly in the weak coverage grades (Kāśikāvṛtti 0/20, Śatapathabrāhmaṇa 15/20). Among the resolved loci, 69 of 74 = 93 % attest the cited reading at the cited locus. The remaining five are individually diagnosable as edition differences.

- **Two snapshots only.** The 2021/2026 comparison (§5.4, §6.2) establishes a direction of change for that interval, not a growth rate or law.

Validation has two layers with different evidentiary reach. Layer 1 tests internal stability across the measured letter-*a* material: fascicle coverage ranges from 16.91 % to 36.90 %, with 12 of 24 measured fascicles outside a ±2-percentage-point band around the headline. The pooled 25.2 % therefore retains its original basis but is not homogeneous across fascicles. Layer 2 would test print–digital representativeness on the print-only fascicles; it stopped at `STOP_NO_PRINT_ACCESS`, so no print validation is claimed. These layers do not remove the permanent whole-alphabet limitation: the headline remains a measurement of the exercised letter-*a* citation-source inventory, not of PD as a whole. Measurement: `https://github.com/sanskrit-lexicon/csl-atlas/blob/main/reports/PD_TWO_LAYER_VALIDATION_2026-10-26.md`.

Three observations would revise the argument. A recount that changes the covered mass would revise the 25.2 % any-presence and 23.3 % strict-sensitivity results. A complete join of the declared bibliography to DCS would replace the 8.0 % observed lower bound with a measured overlap. A print-only fascicle or a second letter with materially different citation composition would restrict generalisation beyond the present letter-*a* census. None of these outcomes would change the arithmetic of the census already measured.

8 Conclusion

On the shared span, PD's letter-*a* citation density is 23.2× that of the finished PWG. Against PD's letter-*a* citation-source inventory, DCS is **deep, not broad**. It holds the Vedic-epic-śāstric spine that carries more than three-quarters of its own token mass and a quarter of the dictionary's citation practice. The purāṇic, lexicographic, and classical-kāvya breadth remains outside it: some two thousand three hundred works and three-quarters of the citation mass. The gap has structure, a ranked residue, and a measured change between the 2021 and 2026 snapshots. The method reads a historical dictionary's citation apparatus as a frequency-salient citation-source inventory and canon proxy, then joins it against a corpus inventory. It is portable to any language with a historical-principles dictionary and a digital corpus; the asymmetric-anchor design (§4.3) makes it tractable even when the apparatus runs to thousands of sigla.

The letter *a* is a masterpiece; the plan was a cathedral, and what stands is a chapel that has been called a foundation.

A The canon behind the denominator

Metric 2 divides by a list of books, and that list has a history. Read closely, PD's canon is *one canon seen through two catalogues* and *two edition layers*. The catalogues are Aufrecht's *Catalogus Catalogorum* (ACC, 1891–1903) and the *New Catalogus Catalogorum* (NCC, 1949–2019): largely the same works, registered twice, a century apart. The edition layers are the editions the two projects cite. The PWG bibliography's dated entries all fall at or before 1870; PD's declared list is dated differently: four entries in five whose OCR tail carries a datable imprint year postdate 1903, the year ACC's last part appeared (interim, AOB denominator).[7] The century contrast therefore belongs to the *edition layer*, not to the works: there is no "ACC is the nineteenth-century canon, NCC the twentieth-century canon" headline to be had from the work titles alone.

Two denominators must never be mixed. (i) The *declared canon* is the printed list of books, once the OCR is corrected against the printed pages (§5.2, §7). (ii) The *letter*-a *citation mass* is what PD actually exercises in the machine-readable text. The first answers "how much of the declared list does DCS hold?"; the second answers "how much of what PD cites does DCS hold?". A ratio may not take its numerator from one denominator and its denominator from the other, and a floor is never a denominator. For the same reason the clean-filter rows of `pd_aob_declared_works.tsv` are a floor, and the partial manuscript index is not a census.

The join behind the table first matches each PD display title against the ACC headwords. It folds away aspiration, retroflexion and vowel length on both sides and drops parentheticals. It then follows only the confirmed Tier A+B rows of the published ACC–NCC crosswalk.

Step	Works	Occurrences	Basis
PD hand-adjudicated primary titles	164	123,453	*interim, AOB denominator*
ACC exact headword match	122 (74.4 %)	106,408 (86.2 %)	*exact headword; interim, AOB denominator*
No exact ACC headword	42 (25.6 %)	17,045 (13.8 %)	*interim, AOB denominator*
Composed PD→ACC→NCC (confirmed Tier A+B)	106 (64.6 %)	87,360 (70.8 %)	*Tier A+B; interim, AOB denominator*
ACC-matched, no confirmed ACC↔NCC row	16	19,048	*no confirmed Tier A+B; interim, AOB denominator*
Direct PD→NCC path (exact NCC headword)	127 (77.4 %)	93,934 (76.1 %)	*exact NCC headword; interim, AOB denominator*
NCC manuscript census	not published	n/a	pending census

Table 4: Interim composition of PD's declared canon through the two manuscript catalogues. Works and occurrences are those of the hand-adjudicated letter-*a* primary titles; percentages are of the first row. Every count is interim, AOB denominator.

[7] Imprint years as recorded per entry in `pd_aob_declared_entries.tsv` (`data/pd/`, csl-atlas); undated entries are excluded. The census is preliminary and is not used as a denominator in this paper.

The composed row intersects the exact-headword join with the confirmed Tier A+B rows of the ACC–NCC crosswalk. Most hand-adjudicated titles acquire at least one confirmed NCC row. The remainder has *no confirmed* row, which is not evidence that NCC lacks the work (Table 4 gives both counts). The direct path is measured too, on the same titles: an exact folded NCC headword match covers three-quarters of them. It is a title match, not yet the siglum join the printed Index makes possible, and it is not additive with the composed row; a work can satisfy both. C/D candidates never enter this count: they may bridge only as `cd_bridge` entries, under a similarity threshold of 0.90, a one-sided PD match and a contradiction check, in a separate ledger.

Beyond the hand-adjudicated titles, the same join run over the full AOB-title layer covers about a third of that layer's citation mass, over some three thousand distinct ACC↔NCC edges (interim, AOB denominator). Its provenance, the C/D bridges (thirty-six of them, fourteen rejected at the 0.90 threshold) and the per-row basis sit in the committed tables. That layer is OCR-grade and largely absent from both catalogues, so its share is context, never a headline, and it is not comparable with the 164-title control.

The composition is provisional in a specific sense. The direct path above is a title match; the siglum join that the printed *Index of Works* (pp. LXXXIX–CXXVII) makes possible is a separate re-run, and the NCC manuscript census is not reported. The Index is now transcribed in full; it prints the AOB sigla in its fifth column, so the recomposition is a join, not a new identification problem. Until that siglum-denominator recomposition lands, every figure in this subsection remains interim, AOB denominator.

B PD and PWG on the same unit

This appendix carries the comparison the body only points to: PD's density against the finished PWG, and the arithmetic behind the claim that PD is three times the OED.

The Sanskrit league makes the same point in years. Wilson took about thirteen years for forty-four thousand lemmas. The *grosses Petersburger Wörterbuch* (PWG: Böhtlingk–Roth, 1855–1875) took twenty years for a hundred and six thousand. Böhtlingk's single-handed abridgement (PWK: *Sanskrit-Wörterbuch in kürzerer Fassung*, 1879–1889) took ten years for a hundred and fifty-one thousand. Monier-Williams took about twenty-seven years for a hundred and ninety-four thousand. None ran past twenty-seven; terseness finishes, exhaustiveness drowns. (This paper writes PWG and PWK throughout, since bare *PW* names the great dictionary in much of the Sanskritist literature but codes the *kürzere* in Cologne's digitisation.)

The two dictionaries can be compared on the same unit because one quotation rule applies to both texts. A quotation is a maximal run of Sanskrit blocks closed by a citation marker; further loci after a semicolon count as attestations. The unit is defined in the §4.2 footnote, and the rule and both measurements are in the `https://github.com/sanskrit-lexicon/csl-atlas/blob/main/reports/PD_QUOTATION_UNIT_2026-10-18.md`. PD's claim is three times the OED. Four ledgers test it, in the order in which each explains the next.

By volumes the claim is arithmetically true: three times over. By pages it is under twice, and by entries a little over twice. The three-ness lives in the counting. It is three times the OED the way a man is three times as tall if you count his hats.

PD has not overgrown PWG in total mass. Inside letter *a* it carries a little over half of PWG's quotation-linked attestations and nine-tenths of the quotations the finished PWG carries across the whole alphabet; PWG still carries more attestations overall. PD has overgrown PWG where it has gone, per headword: twenty-three times its density in the shared span. This is a directional ratio, lemmas against headwords, and a lemmatisation choice, not a citation measure: seven headwords in ten are compounds, one in eight a derivative and one in six simplex, with homonyms adding a fortieth of the entries. The median shared lemma has four times the attestations, not the twelvefold of the single entry Wright chose; one shared lemma in twelve

Ledger	PWG	PD
Sources declared	502 bibliography entries	1,476 AOB rows: 2.94× PWG, 3.3 % of the 44,773-work crosswalk
Ground covered	complete alphabet	*a–apaca-*, 4.3 % of PWG's alphabetical ground
Output inside the ground	106,082 headwords · 729,719 quotation-linked attestations (801,788 reference tags) · 329,623 quotations	104,959 lemmas: 99 % of PWG's headword count, not 99 % coverage · 419,890 attestations (57.5 % of the quotation-linked mass; 52.4 % of the tags) · 302,442 quotations (91.8 %)
Outcome	20 years, complete	50 years of printing, 4.3 % of the alphabet, completion ≈2284–3211

Table 5: The four ledgers behind PD's "three times the OED" claim, in the order in which each explains the next. Bibliographic entries, not works; the quotation/attestation unit is defined in §4.2.

falls below parity.[8]

The rate explains the outcome. PD produced about two thousand lemmas a year against PWG's five thousand, with a hundred and twenty printed pages a year and seventy attestations per printed page. The plan was kept in volumes, the work in articles, and the supply in slips; they never met.

Three families show the policy at work, the third as its exception: PD gives a lemma to nearly every compound and productive derivative, PWG to a selection, except among function words.

Family	PD headwords	PWG headwords	PD compounds	PD derivatives	PD simplex
anna	877	33	77.5 %	8.7 %	13.8 %
agni	2,312	170	73.7 %	14.9 %	11.4 %
atha	173	24	63.1 %	7.5 %	29.4 %

Table 6: Two letter-*a* families under the two lemmatisation policies: PD heads a lemma for nearly every compound and productive derivative, PWG for a selection. Percentages are of the classified headwords in each family.

The privative *a-* and the negative *an-* account for about one headword in eleven; without them the density multiple falls from twenty-three to twenty-one, still an order of magnitude. (The accent-blind count, in which Vedic accent marks hide the privatives, gives the same picture: `https://github.com/sanskrit-lexicon/csl-atlas/blob/main/reports/PD_FAMILY_COMPOSITION_2026.md`.)

Acknowledgements

I thank Nāgabhūṣaṇarāvu Kālepu for scrupulous re-examination of hastily OCRed dictionary apparatus, a standing reminder that what a dictionary text recovers depends on how patiently it was read.

AI tools were used in the production of this paper, and their scope is stated here rather than left to the submission form. Agent sessions drafted and ran the extraction, join, figure, and verification scripts (DeepSeek Flash `deepseek-flash`; GLM-5.3 `zai-coding-plan/glm-5.3-flash`; Claude Fable 5 `claude-fable-5`). The same tools drafted the English prose, which the author then edited. Every headline number is re-derived from the committed public data by a script that aborts on drift, and every citation was checked to resolve. The author set the argument, edited the prose, and accepts responsibility for the text.

[8] Per lemma overall, the average PD headword is *less* cited than PWG's, because the hundred thousand PD-only lemmas are thin compounds. The multiples hold only on the shared lemmas. There PD carries about nine times the quotations and nine times the attestations (pending a per-lemma re-run).

Data availability

All inputs and figures are public. Derived tables and figures are in the `csl-atlas` repository (`data/pd/`, `scripts/`, `papers/figures/`). The PD source text is the Cologne PDScan 2020 edition (© The Sanskrit Library / Thomas Malten, CC BY-NC-SA 3.0); DCS token inventories come from VisualDCS and the DCS abbreviation list. The composition in Appendix A uses `pd_siglum_families.tsv` and `pd_aob_declared_entries.tsv` (csl-atlas, `data/pd/`), together with `acc.jsonl` and `works_crosswalk.tsv` (SanskritLexicography, `HeadwordLists/works_catalogue/`). It applies the folding rule stated there, so its counts can be re-derived from those tables. Figures regenerate deterministically with `python scripts/build_a68_figures.py`; a standalone canary re-checks all 12 headline values against the committed report.

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